instruments, including radar. Because *Cassini* will be so much closer to Titan, it will be able to resolve many details on Titan's surface indetectable to Muhleman's pioneering Earth-based system. It's also likely that the surface can be viewed in the near infrared. Maps of the hidden surface of Titan may be in our hands sometime in the summer of 2004.

Cassini is also carrying an entry probe, fittingly called *Huygens*, which will detach itself from the main spacecraft and plummet into Titan's atmosphere. A great parachute will be deployed. The instrument package will slowly settle through the organic haze down into the lower atmosphere, through the methane clouds. It will examine organic chemistry as it descends, and—if it survives the landing—on the surface of this world as well.

Nothing is guaranteed. But the mission is technically feasible, hardware is being built, an impressive coterie of specialists, including many young European scientists, are hard at work on it, and all the nations responsible seem committed to the project. Perhaps it will actually come about. Perhaps winging across the billion miles of intervening interplanetary space will be, in the not too distant future, news about how far along the path to life Titan has come.

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*There could have been none. We're very lucky that there *is* such a world to study. The others all have too much hydrogen, or not enough, or no atmosphere at all.

*Not because he thought it remarkably large, but because in Greek mythology members of the generation preceding the Olympian gods—Saturn, his siblings, and his cousins—were called Titans.

*Titan's atmosphere has no detectable oxygen, so methane is not wildly out of chemical equilibrium—as it is on Earth—and its presence is in no way a sign of life.